27-6 Yahrzeit Wall Overview

The Yahrzeit Panels are etched glass panels with an automated lighting mechanism. These panels memorialize the names and dates of death of members of the congregation or their families.

In observance of individual Yahrzeit and Yizkor dates, memorial LEDs are lit and extinguished according to programmed synagogue policy, typically at the start of the Shabbat preceding the Yahrzeit. Yahrzeit and Yizkor dates may be based on either the Jewish or the secular (Gregorian) calendar. The memorial LED lighting time may be scheduled automatically, triggered manually, based on sunset, or set to a fixed time.

Manufacturer:	On-Site Systems Inc.	http://www.on-sitesystems.com	
	Yyz Systems Inc	http://www.yyzsystems.com	
	Allan Schwartz	(server and controller)	
Model Number:	*custom installation*		
Maintenance:	Clean the glass monthly, with Windex wipes or equivalent		
Service:	Glass – On-Site Systems		
	Electronics – Yyz Systems		
	Software – Allan Schwartz		
	Repair/reprint panels of glass – On-Site Systems		
Location:	Sanctuary west wall		
Installers:	On-Site Systems Inc.	http://www.on-sitesystems.com	
	Gary Taylor	gary@on-sitesystems.com	800-838-7778
	Yyz Systems Inc	http://www.yyzsystems.com	
	Marko Hajdinovic	marko@yyzsystems.com	905-795-8500
	Allan Schwartz	allanschwartz@sbcglobal.net	669-350-7296

Physical Wall Overview

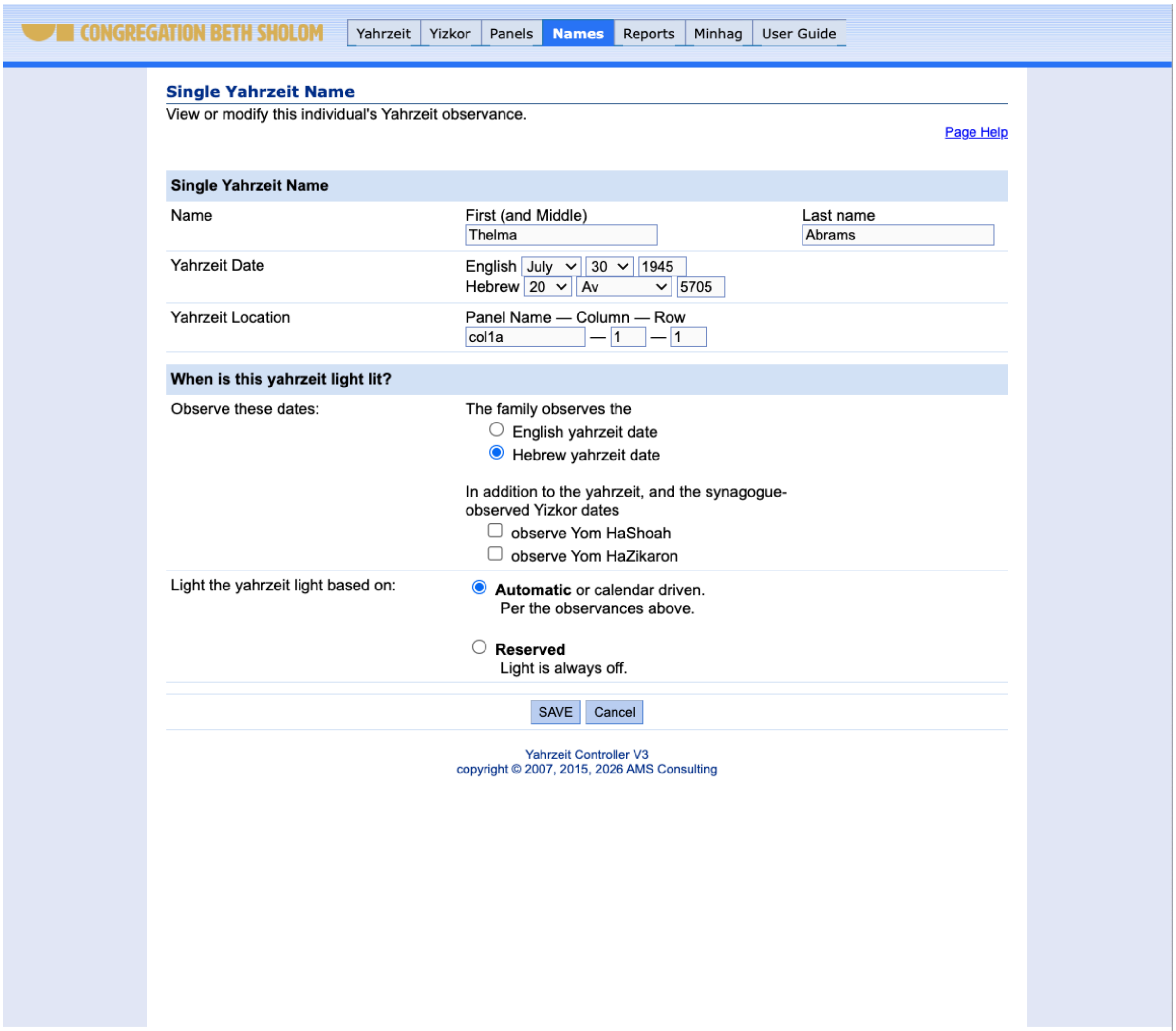
The Yahrzeit Wall comprises the Glass Panels (see section **27-7 Yahrzeit Panels**), the embedded electronics in the wall, and the Yahrzeit Server software running on the Yahrzeit appliance or server (see section **27-8 Yahrzeit Server**).

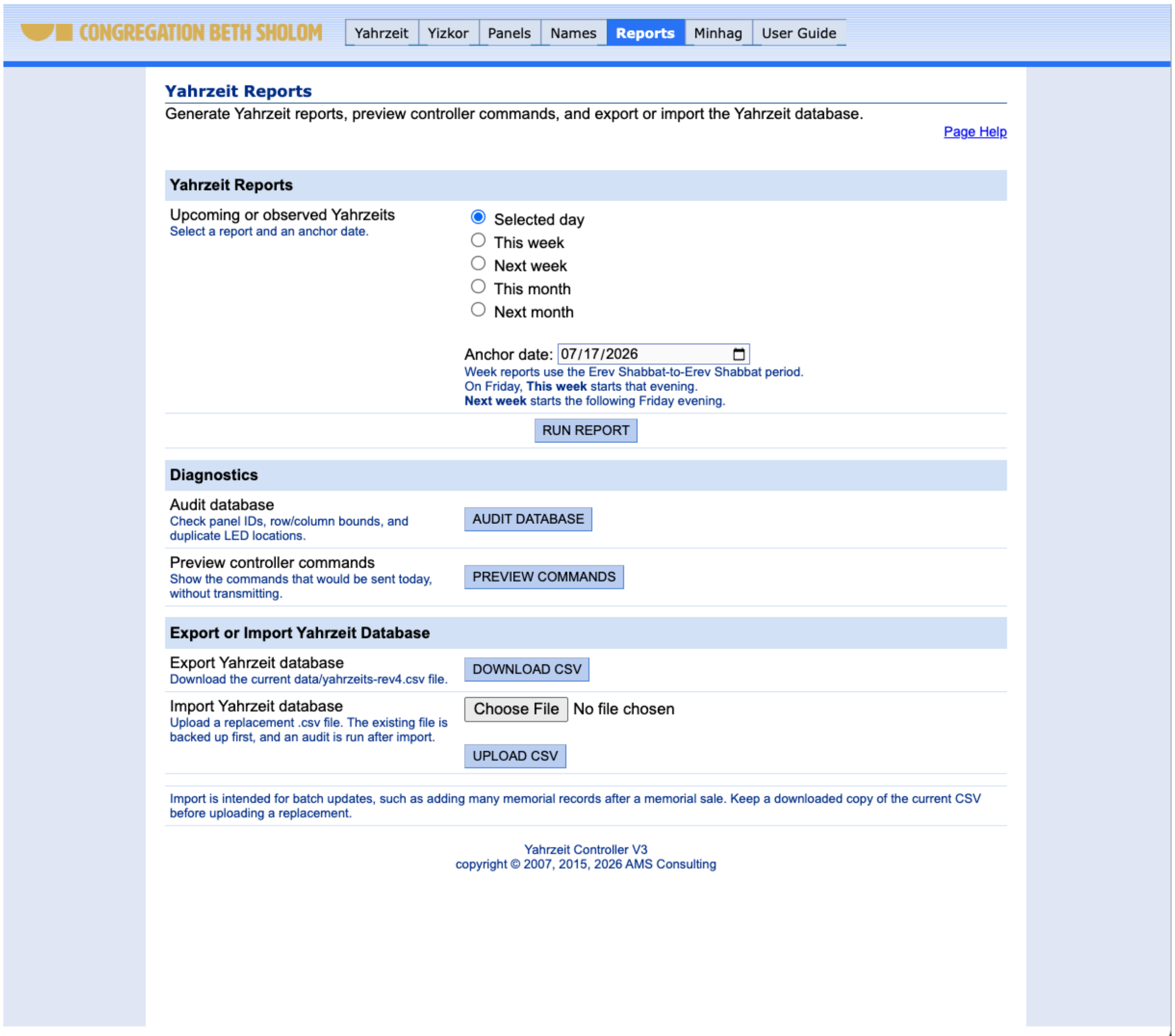
This document doesn't discuss how the software operates, or is layered, or how the end-user interfaces with this software. See section **27-8 Yahrzeit Server.**

This document doesn't discuss the glass, or how to create or modify the panels. See section **27-7 Yahrzeit Panels.**

Purpose of the System

The Yahrzeit System provides a browser-based interface for maintaining memorial records, configuring synagogue lighting policy, checking wall status, and operating the calendar-based Yahrzeit and Yizkor lighting schedule. The user interface is a set of browser pages, such as:





CONGREGATION BETH SHOLOM

Yahrzeit | Yizkor | Panels | Names | **Reports** | Minhag | User Guide

Yahrzeit Reports

Generate Yahrzeit reports, preview controller commands, and export or import the Yahrzeit database.

Page Help

Yahrzeit Reports

Upcoming or observed Yahrzeits
Select a report and an anchor date.

- (●) Selected day
- ( ) This week
- ( ) Next week
- ( ) This month
- ( ) Next month

Anchor date: 07/17/2026

Week reports use the Erev Shabbat-to-Erev Shabbat period.
On Friday, **This week** starts that evening.
Next week starts the following Friday evening.

RUN REPORT

Diagnostics

Audit database
Check panel IDs, row/column bounds, and duplicate LED locations.

AUDIT DATABASE

Preview controller commands
Show the commands that would be sent today, without transmitting.

PREVIEW COMMANDS

Export or Import Yahrzeit Database

Export Yahrzeit database
Download the current data/yahrzeits-rev4.csv file.

DOWNLOAD CSV

Import Yahrzeit database
Upload a replacement .csv file. The existing file is backed up first, and an audit is run after import.

Choose File No file chosen

UPLOAD CSV

Import is intended for batch updates, such as adding many memorial records after a memorial sale. Keep a downloaded copy of the current CSV before uploading a replacement.

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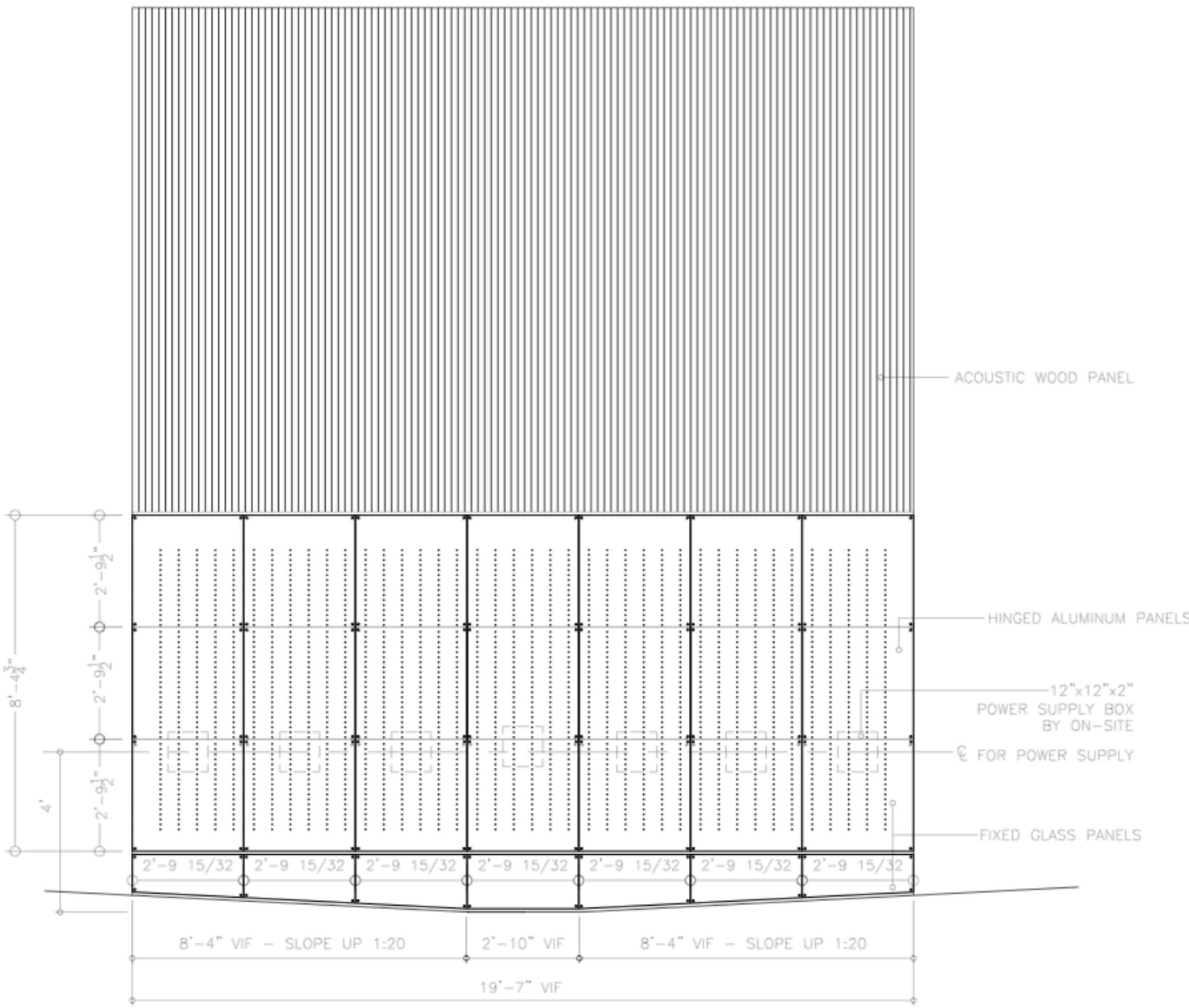
The Yahrzeit Wall Structure



The Yahrzeit Wall Structure consists of seven large aluminum panels hinged at the top. Each aluminum panel supports three glass panels and has drilled openings through which the LEDs are visible. Circuit boards containing LEDs and supporting circuitry are mounted to the inside of the aluminum panels. Power supplies mounted inside the wall feed the LED circuitry.

The Yahrzeit Controller is an embedded microprocessor controller located behind the first large aluminum panel. It interfaces with the Yahrzeit LED Array and controls which memorial LEDs are lit.

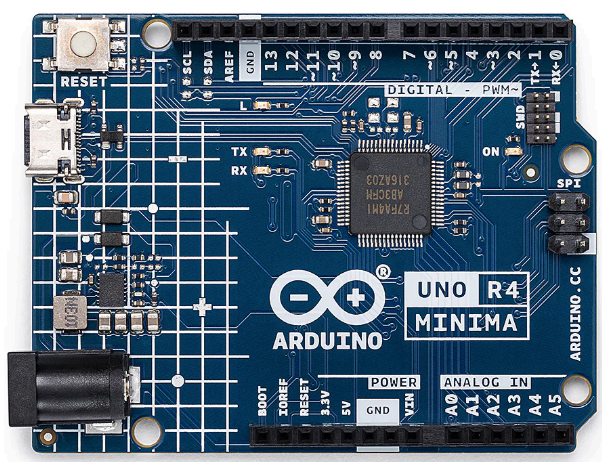
The Yahrzeit Server resides in the synagogue office. It provides the browser-based user interface and runs the background processes that automate the Yahrzeit panels.

Yahrzeit Wall Components

This section briefly defines the components that make up the integrated Yahrzeit Wall.

Yahrzeit Panels –The twenty-one etched glass panels provide space for up to 2,240 names to be remembered.

Yahrzeit Server – A Linux-based appliance that presents the browser user interface, runs the background processes that perform Yahrzeit and Yizkor automation, and signals the Yahrzeit Controller to change the LED display..

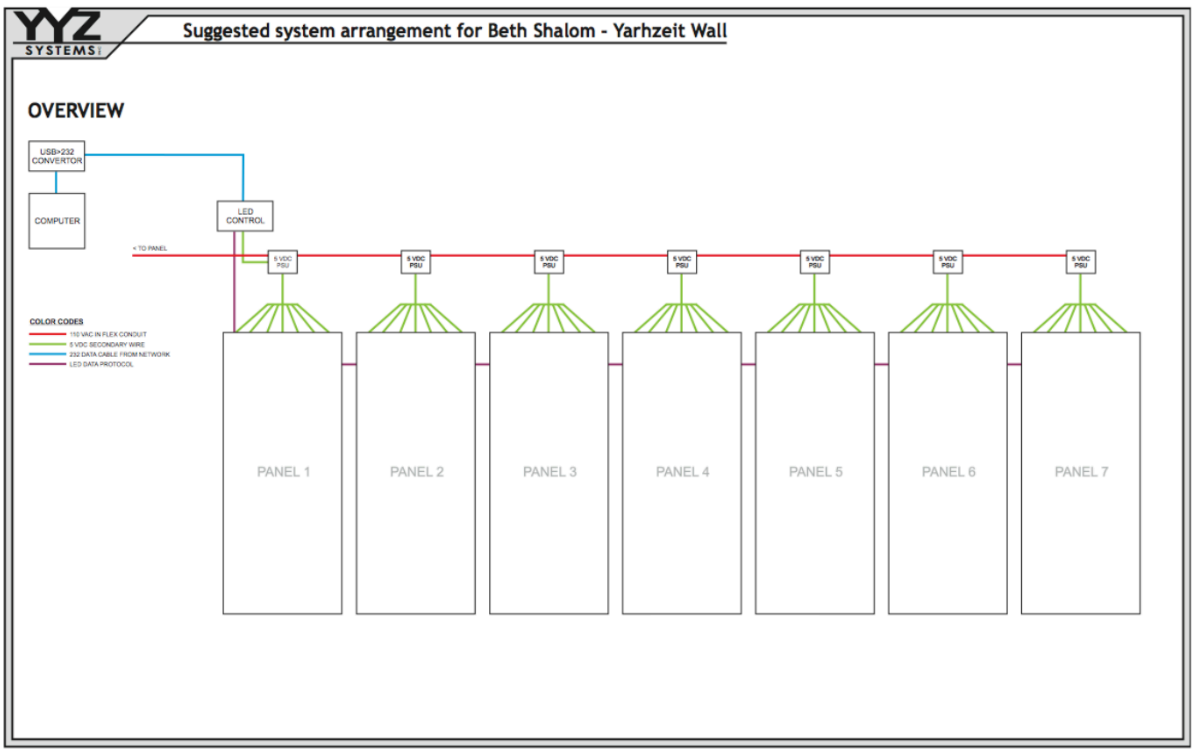


Yahrzeit Controller – The Arduino-based embedded controller installed behind the Yahrzeit Wall. It receives commands from the Yahrzeit Server and drives the Yahrzeit LED Array by refreshing the pixel boards with the current LED state.

Yahrzeit LED Array – The 56-row by 40-column LED array, for a total of 2,240 individual memorial LEDs.

Yahrzeit Pixel board (YPX) – The YPX board is a small number (6 or 8 or 10) of pixels or LEDs on a single printed circuit board. A string of 7 YPX boards comprise each column of 56 LEDs and 40 columns – a total of 280 YPX boards – are connected forming the entire LED array.

Interconnect cables – The Yahrzeit Pixel boards are interconnected in strings of seven YPX boards for each 56-LED column. Separate power interconnects distribute power to the pixel boards.



Yahrzeit Controller

The current Yahrzeit Controller is an Arduino Uno R4 Minima stack with an Ethernet shield and a custom pixel-interface shield. It replaces the original v1 8051-based controller, the separate serial-to-Ethernet bridge used in the original installation, and the v2 Arduino Mega controller used in the 2015 refresh.

The Yahrzeit Server communicates with the controller over Ethernet using a simple text command protocol. The controller maintains the LED wall state, maps logical wall/panel positions to the physical pixel chain, and refreshes the LED array behind the glass panels.

During normal operation, synagogue staff do not need to interact with the controller directly. Controller diagnostics and firmware maintenance are engineering/service tasks.

Yahrzeit Server / Appliance System

There are three layers of Yahrzeit software:

- The browser-based management software, used by synagogue staff to maintain memorial records, configure synagogue policy, view status, and run reports. This runs on the Yahrzeit Server / Appliance and is based on the standard Linux/Apache/PHP web application stack.
- The background automation software, which runs scheduled Yahrzeit and Yizkor operations and sends the required display commands to the controller. This software runs automatically during normal operation.
- The embedded controller firmware, which runs on the Arduino-based Yahrzeit Controller. This firmware maintains the LED wall state and drives the LED array at the hardware level.

The current Yahrzeit Server / Appliance is a dedicated Linux-based computer located in the synagogue office or other designated equipment location. See section **27-8 Yahrzeit Server**, for the appliance address, login/service notes, backups, scheduled operation, and maintenance procedures.

The Yahrzeit Server's Browser User Interface

Normal operation of the Yahrzeit Server is through its browser-based user interface. Synagogue staff use these pages to view wall status, maintain memorial records, configure synagogue policy, and run reports or controlled wall operations.

The complete user interface is described in section **27-8 Yahrzeit Server**. The installed browser address should be recorded there after the server/appliance is configured on the synagogue network.

The sample pages below show the wall status/controller page and the Minhag configuration page.

- **Yahrzeit Controller**: This page summarizes the current state of the Yahrzeit Wall and provides links to related status, configuration, and maintenance pages.

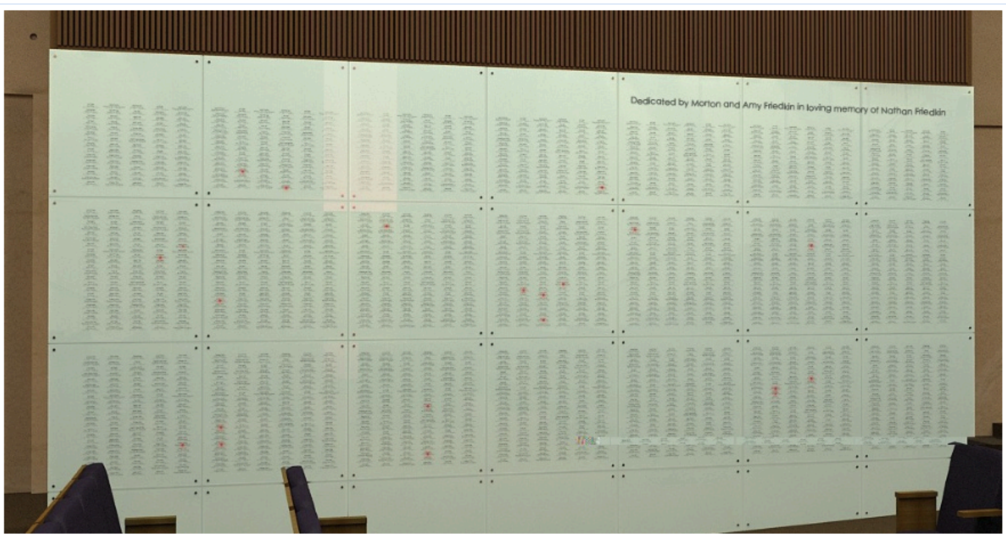
CONGREGATION BETH SHOLOM | Yahrzeit | Yizkor | Panels | Names | Reports | Minhag | User Guide

Yahrzeit Controller

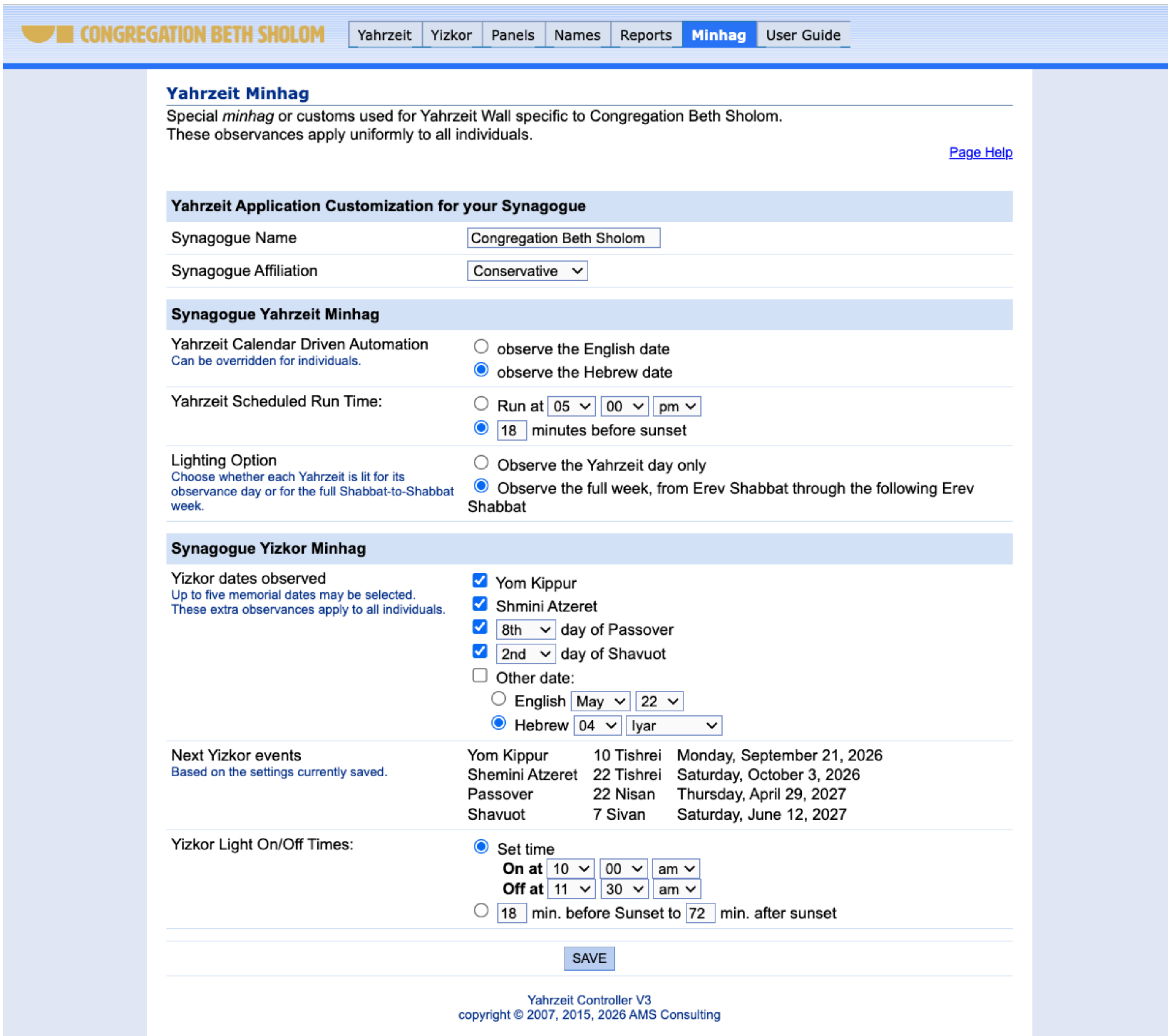
Home and status page for the Yahrzeit panels at Congregation Beth Sholom.

Page Help

Congregation Beth Sholom Yahrzeit Controller	
Date / Time	Friday July 17, 2026, 12:55 pm 3 Av 5786
Addresses	Yahrzeit server: unknown Yahrzeit controller: 192.168.86.240
Scheduled Events	Today's sunset in San Francisco is at 8:30 pm. On Friday July 17, 2026, 8:30 pm this week's yahrzeits will be lit.
Configured Policy Summary	Yahrzeits are normally observed by Hebrew date Yahrzeit lighting uses a sunset-relative window Yizkor lighting uses a fixed clock time Minhag settings are edited on the **Minhag** screen
Controller Summary	21 panels defined (click on **Panels**) 1411 names defined (click on **Names**) 24 memorial lights are lit now. Reports, audits, command previews, and CSV maintenance are available on the **Reports** screen Manual lighting operations are available from the **Yizkor** screen



- **Minhag**: This page configures the synagogue-specific customs and observances used by the Yahrzeit and Yizkor lighting schedule.



CONGREGATION BETH SHOLOM

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Yahrzeit Minhag

Special *minhag* or customs used for Yahrzeit Wall specific to Congregation Beth Sholom.
These observances apply uniformly to all individuals.

Page Help

Yahrzeit Application Customization for your Synagogue	
Synagogue Name	Congregation Beth Sholom
Synagogue Affiliation	Conservative
Synagogue Yahrzeit Minhag	
Yahrzeit Calendar Driven Automation Can be overridden for individuals.	○ observe the English date ◉ observe the Hebrew date
Yahrzeit Scheduled Run Time:	○ Run at 05 : 00 pm ◉ 18 minutes before sunset
Lighting Option Choose whether each Yahrzeit is lit for its observance day or for the full Shabbat-to-Shabbat week.	○ Observe the Yahrzeit day only ◉ Observe the full week, from Erev Shabbat through the following Erev Shabbat
Synagogue Yizkor Minhag	
Yizkor dates observed Up to five memorial dates may be selected. These extra observances apply to all individuals.	☑ Yom Kippur ☑ Shmini Atzeret ☑ 8th day of Passover ☑ 2nd day of Shavuot ☐ Other date: ○ English May 22 ◉ Hebrew 04 Iyar
Next Yizkor events Based on the settings currently saved.	Yom Kippur 10 Tishrei Monday, September 21, 2026 Shemini Atzeret 22 Tishrei Saturday, October 3, 2026 Passover 22 Nisan Thursday, April 29, 2027 Shavuot 7 Sivan Saturday, June 12, 2027
Yizkor Light On/Off Times:	◉ Set time **On at** 10 : 00 am **Off at** 11 : 30 am ○ 18 min. before Sunset to 72 min. after sunset

SAVE

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Related Documentation

This overview describes the purpose and major components of the Yahrzeit Wall. More detailed information is maintained in the following House Manual sections:

- **27-7 Yahrzeit Panels** – glass panels, panel layout, physical panel maintenance, and panel repair/reprint information.
- **27-8 Yahrzeit Server** – server/appliance operation, browser access, scheduled operation, backups, network addresses, and maintenance procedures.
- **27-3 Low Voltage Room** – building electronics location and infrastructure, where applicable.
- **27-4 Networking** – building network information, where applicable.

Engineering source files, firmware, hardware drawings, and appliance software are maintained in the public Yahrzeit source repository on GitHub. The installed appliance normally contains a sparse checkout under `/home/allan/src/yahrzeit`, with the web appliance in `/home/allan/src/yahrzeit/yahrzeit_site-v3`. The full project tree is available from the public GitHub repository referenced by that checkout. These files are intended for service, repair, refresh, and future engineering work, not for normal synagogue staff operation.